COMPUTER NETWORKS
PROJECT REPORT

Branch-to-Headquarters Network Design
with OSPF, RIP and Route Redistribution

Table of Contents

1. Introduction
2. Network Topology Overview
3. Device and IP Addressing Table
4. VLAN Configuration
5. Router Configuration
6. OSPF Configuration
7. RIP Configuration
8. Route Redistribution
9. DHCP Configuration
10. DNS and Web Server
11. Verification and Testing
12. Troubleshooting - Routing Loop
13. Conclusion

1. Introduction

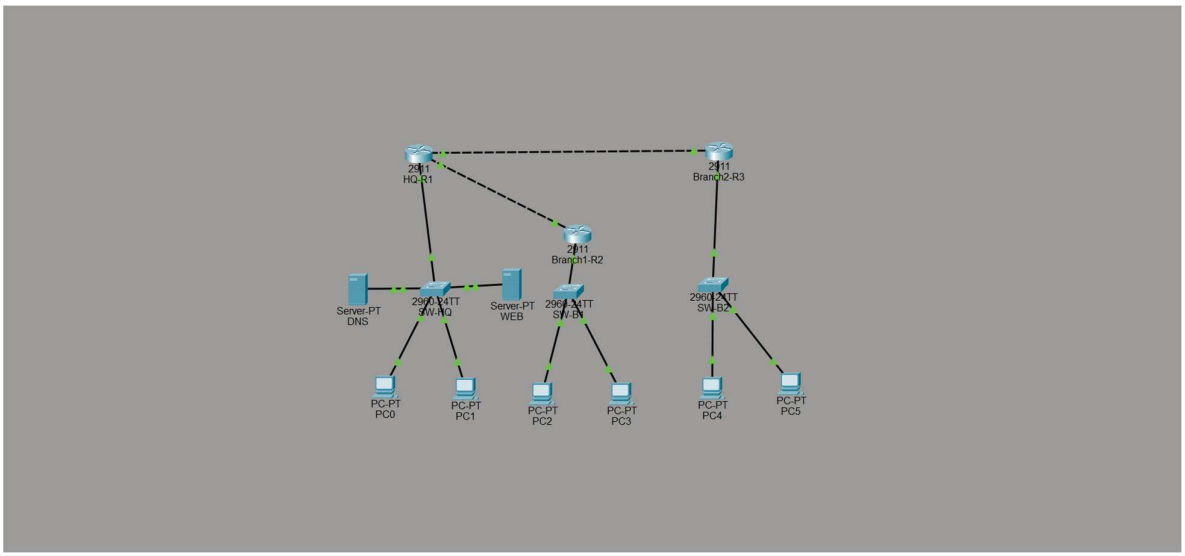
This project demonstrates the design and implementation of a branch-to-headquarters enterprise network using Cisco Packet Tracer. The network connects three sites: one Headquarters (HQ) and two Branch offices, using two different routing protocols — OSPF and RIP — with route redistribution at the HQ router.

The project covers the following key areas:

- Multi-site network topology design with 3 routers
- VLAN configuration at each site
- OSPF routing protocol for HQ and Branch2
- RIP routing protocol for Branch1
- Route redistribution between OSPF and RIP at HQ-R1
- DHCP configured on routers for automatic IP assignment
- DNS and Web server setup and testing
- Routing loop identification and troubleshooting

2. Network Topology Overview

The network consists of three routers interconnected to form a branch-to-headquarters topology. HQ-R1 acts as the central border router connecting both branch sites.



2.1 Topology Description

Device	Role	Protocol	Connected To
HQ-R1	Headquarters Main Router / Border Router	OSPF + RIP + Redistribution	Branch1-R2, Branch2-R3, SW-HQ
Branch1-R2	Branch 1 Router	RIP Only	HQ-R1, SW-B1
Branch2-R3	Branch 2 Router	OSPF Only	HQ-R1, SW-B2
SW-HQ	HQ Switch	VLAN 10, 20	HQ-R1, DNS, WEB,PC0, PC1
SW-B1	Branch1 Switch	VLAN 10, 20	Branch1-R2, PC2, PC3
SW-B2	Branch2 Switch	VLAN 10, 20	Branch2-R3, PC4, PC5

3. Device and IP Addressing Table

Device	Interface	IP Address	Subnet Mask	Description
HQ-R1	Gi0/0.10	192.168.0.1	255.255.255.0	HQ VLAN 10
HQ-R1	Gi0/0.20	192.168.1.1	255.255.255.0	HQ VLAN 20
HQ-R1	Gi0/1	10.0.0.1	255.255.255.252	Link to Branch1-R2
HQ-R1	Gi0/2	10.0.0.5	255.255.255.252	Link to Branch2-R3
Branch1-R2	Gi0/1	10.0.0.2	255.255.255.252	Link to HQ-R1
Branch1-R2	Gi0/0.10	192.168.10.1	255.255.255.0	Branch1 VLAN 10
Branch1-R2	Gi0/0.20	192.168.11.1	255.255.255.0	Branch1 VLAN 20
Branch2-R3	Gi0/2	10.0.0.6	255.255.255.252	Link to HQ-R1
Branch2-R3	Gi0/0.10	192.168.20.1	255.255.255.0	Branch2 VLAN 10
Branch2-R3	Gi0/0.20	192.168.21.1	255.255.255.0	Branch2 VLAN 20
DNS Server	Fa0	192.168.0.60	255.255.255.0	HQ DNS Server
WEB Server	Fa0	192.168.0.50	255.255.255.0	HQ Web Server

4. VLAN Configuration

VLANs are configured on all switches to segment network traffic at each site. Each site uses VLAN 10 (SALES) and VLAN 20 (IT). Router-on-a-Stick configuration is used on all routers with sub-interfaces.

4.1 HQ Switch (SW-HQ) - show vlan brief

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Fa0/6, Fa0/7, Fa0/8, Fa0/9
                                                Fa0/10, Fa0/11, Fa0/12, Fa0/13
                                                Fa0/14, Fa0/15, Fa0/16, Fa0/17
                                                Fa0/18, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Gig0/1, Gig0/2
10   SALES                            active    Fa0/1, Fa0/3, Fa0/4, Fa0/5
20   IT                               active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

4.2 Branch1 Switch (SW-B1) - show vlan brief

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Gig0/1, Gig0/2
10   SALES                            active    Fa0/1
20   IT                               active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

4.3 Branch2 Switch (SW-B2) - show vlan brief

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Gig0/1, Gig0/2
10   SALES                            active    Fa0/1
20   IT                               active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

4.4 Trunk Interface Verification

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/24      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Fa0/24      1-1005

Port        Vlans allowed and active in management domain
Fa0/24      1,10,20

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/24      1,10,20
```

5. Router Configuration

5.1 HQ-R1 Interface Configuration

```
Router>enable
Router#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  192.168.0.1     YES manual up                    up
GigabitEthernet0/0.20  192.168.1.1     YES manual up                    up
GigabitEthernet0/1     10.0.0.1        YES manual up                    up
GigabitEthernet0/2     10.0.0.5        YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

HQ-R1 uses sub-interfaces for Router-on-a-Stick VLAN routing:

```
interface GigabitEthernet0/0.10
 encapsulation dot1Q 10
 ip address 192.168.0.1 255.255.255.0
interface GigabitEthernet0/0.20
 encapsulation dot1Q 20
 ip address 192.168.1.1 255.255.255.0
```

6. OSPF Configuration

OSPF (Open Shortest Path First) is configured between HQ-R1 and Branch2-R3. OSPF uses link-state routing with Router IDs assigned as 1.1.1.1 (R1), 2.2.2.2 (R2), and 3.3.3.3 (R3).

6.1 OSPF Neighbor Adjacency - HQ-R1

```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
3.3.3.3           1   FULL/DR         00:00:38    10.0.0.6        GigabitEthernet0/2
Router#
```

6.2 OSPF Configuration on HQ-R1

```
router ospf 1
 router-id 1.1.1.1
 network 192.168.0.0 0.0.0.255 area 0
 network 192.168.1.0 0.0.0.255 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
```

Neighbor ID	State	Interface	Address
2.2.2.2 (Branch1-R2)	FULL/BDR	GigabitEthernet0/1	10.0.0.2
3.3.3.3 (Branch2-R3)	FULL/BDR	GigabitEthernet0/2	10.0.0.6

7. RIP Configuration

RIP version 2 (RIPv2) is configured on HQ-R1 and Branch1-R2. RIPv2 supports classless routing and is used for the Branch1 site network.

7.1 RIP Configuration on Branch1-R2

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 24 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/0.10 22
  GigabitEthernet0/0.20 22
  GigabitEthernet0/1    22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
        10.0.0.0
        192.168.10.0
        192.168.11.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
        10.0.0.1             120      00:00:20
Distance: (default is 120)
```

```
router rip
 version 2
 no auto-summary
 network 10.0.0.0
 network 192.168.10.0
 network 192.168.11.0
```

7.2 RIP Database on Branch1-R2

```
Router#show ip protocols

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 3.3.3.3
  It is an autonomous system boundary router
  Redistributing External Routes from,
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.20.0 0.0.0.255 area 0
    192.168.21.0 0.0.0.255 area 0
    10.0.0.4 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:26:05
    3.3.3.3              110      00:26:03
  Distance: (default is 110)
```

8. Route Redistribution

Route redistribution is configured exclusively on HQ-R1, which acts as the Autonomous System Boundary Router (ASBR). This allows OSPF routes to be shared into the RIP domain and RIP routes to be shared into the OSPF domain.

8.1 Redistribution Configuration on HQ-R1

```
router ospf 1
 redistribute rip subnets
!
router rip
 version 2
 distance 90
 passive-interface GigabitEthernet0/2
 redistribute ospf 1 metric 5
```

8.2 HQ-R1 show ip protocols

```
Router#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 14 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip, ospf 1
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  GigabitEthernet0/1    22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
        10.0.0.0
Passive Interface(s):
        GigabitEthernet0/2
Routing Information Sources:
        Gateway         Distance      Last Update
        10.0.0.2              90      00:00:27
Distance: 90 (default is 120)

Routing Protocol is "ospf 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Router ID 1.1.1.1
  It is an autonomous system boundary router
  Redistributing External Routes from,
    rip
  Number of areas in this router is 1. 1 normal 0 stub 0 nssa
  Maximum path: 4
  Routing for Networks:
    192.168.0.0 0.0.0.255 area 0
    192.168.1.0 0.0.0.255 area 0
    192.168.0.0 0.0.1.255 area 0
    10.0.0.0 0.0.0.255 area 0
    10.0.0.0 0.0.0.3 area 0
    10.0.0.4 0.0.0.3 area 0
  Routing Information Sources:
    Gateway         Distance      Last Update
    1.1.1.1              110      00:27:06
    3.3.3.3              110      00:27:05
  Distance: (default is 110)
```

8.3 Final Routing Table - HQ-R1 show ip route

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/2
L       10.0.0.5/32 is directly connected, GigabitEthernet0/2
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.20
R    192.168.10.0/24 [90/1] via 10.0.0.2, 00:00:01, GigabitEthernet0/1
R    192.168.11.0/24 [90/1] via 10.0.0.2, 00:00:01, GigabitEthernet0/1
O    192.168.20.0/24 [110/2] via 10.0.0.6, 00:00:12, GigabitEthernet0/2
O    192.168.21.0/24 [110/2] via 10.0.0.6, 00:00:12, GigabitEthernet0/2
```

The routing table on HQ-R1 confirms successful redistribution:

Route	Protocol	Next Hop	Meaning
192.168.10.0/24	R [90/1]	10.0.0.2 (R2)	Branch1 VLAN10 learned via RIP
192.168.11.0/24	R [90/1]	10.0.0.2 (R2)	Branch1 VLAN20 learned via RIP
192.168.20.0/24	O [110/2]	10.0.0.6 (R3)	Branch2 VLAN10 learned via OSPF
192.168.21.0/24	O [110/2]	10.0.0.6 (R3)	Branch2 VLAN20 learned via OSPF

9. DHCP Configuration

DHCP is configured directly on the routers (no separate DHCP server). Each router provides IP addresses dynamically to PCs in its VLANs.

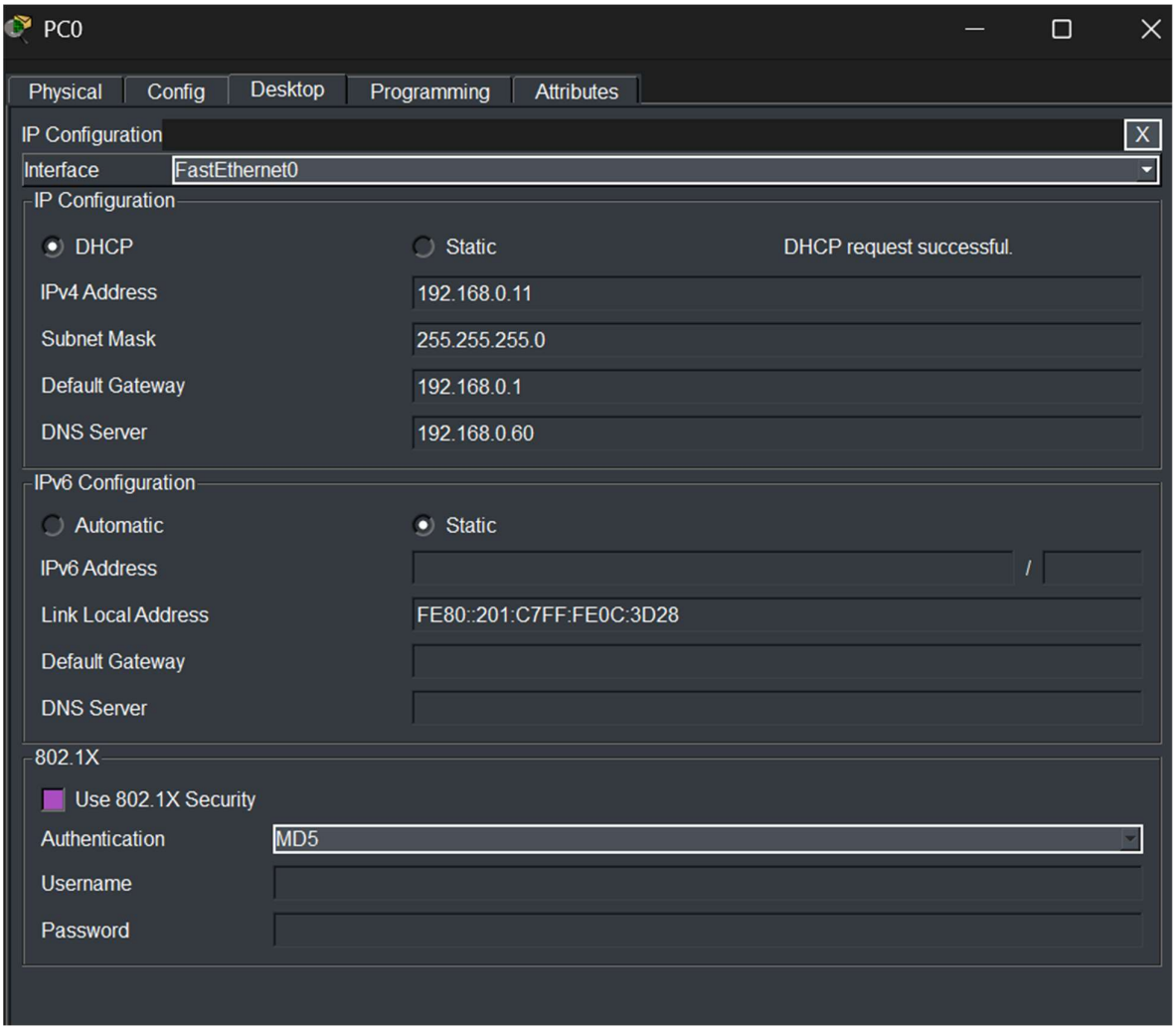
9.1 DHCP Pool Configuration - HQ-R1

```
Router#show ip dhcp pool

Pool VLAN10 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.0.1          192.168.0.1      - 192.168.0.254     1    / 2     / 254

Pool VLAN20 :
 Utilization mark (high/low)    : 100 / 0
 Subnet size (first/next)       : 0 / 0
 Total addresses                : 254
 Leased addresses               : 1
 Excluded addresses             : 2
 Pending event                  : none

 1 subnet is currently in the pool
 Current index        IP address range                    Leased/Excluded/Total
 192.168.1.1          192.168.1.1      - 192.168.1.254     1    / 2     / 254
```

9.2 DHCP Binding Verification

```
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
192.168.0.11     0001.C70C.3D28           --                     Automatic
192.168.1.11     00E0.A3C3.7B27           --                     Automatic
```

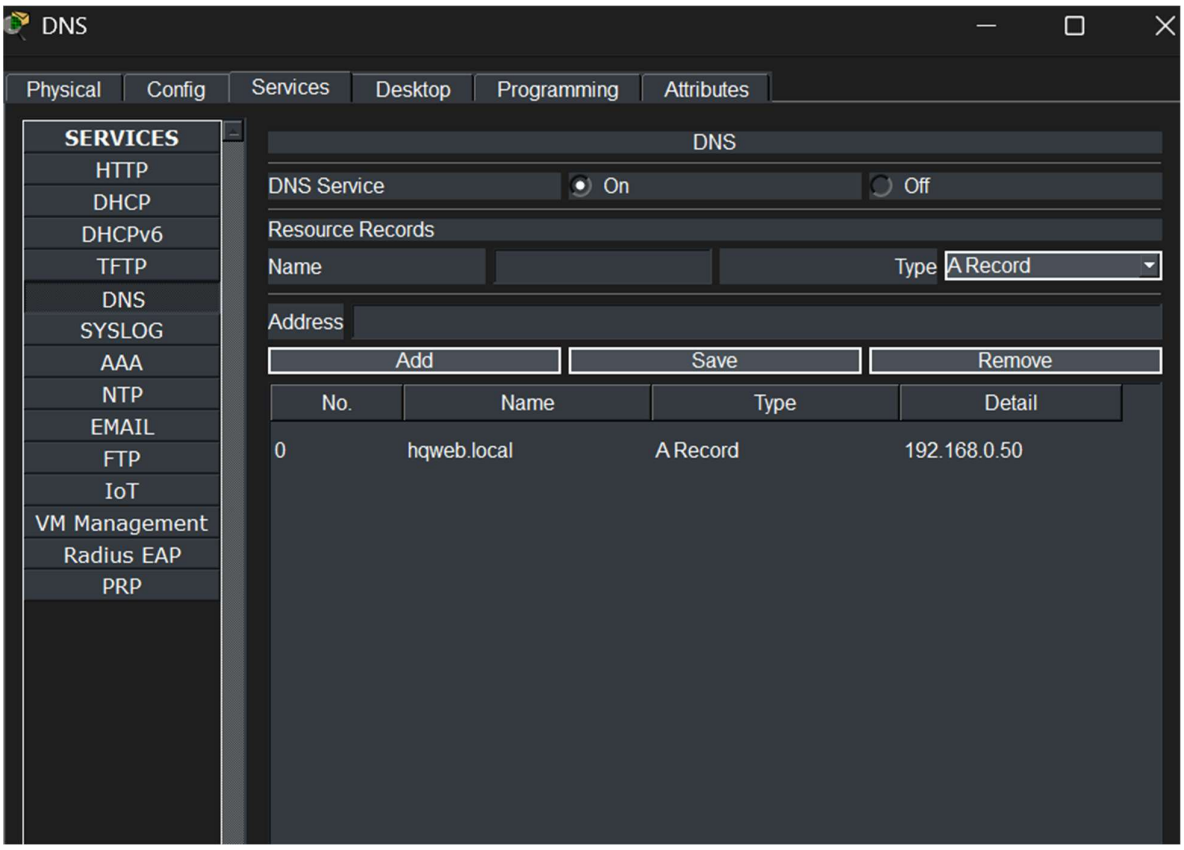
9.3 PC Receiving IP via DHCP



10. DNS and Web Server

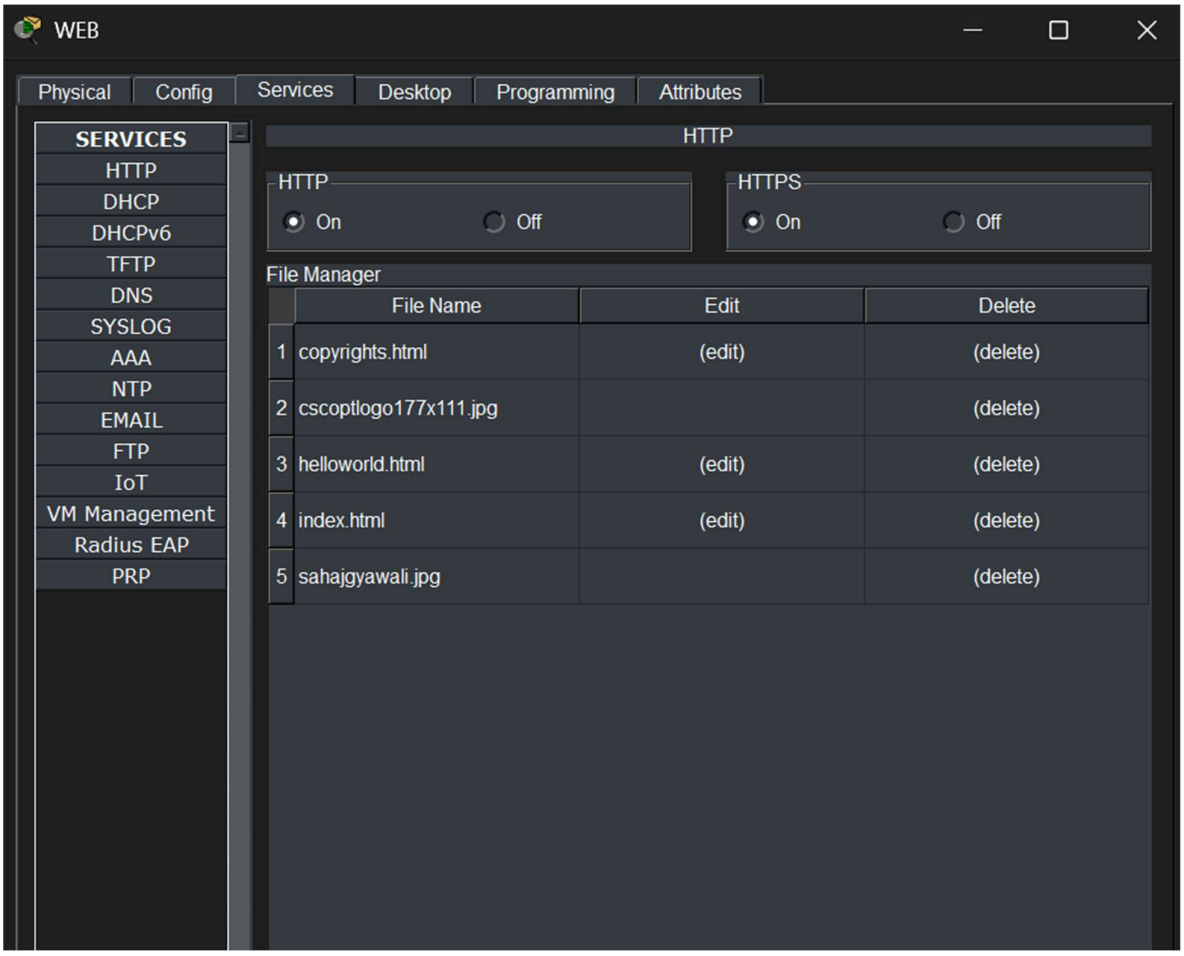
10.1 DNS Server Configuration

The DNS server is located at the HQ site with IP address 192.168.0.60. It resolves the hostname hqweb.local to the Web server IP 192.168.0.50.



10.2 Web Server Configuration

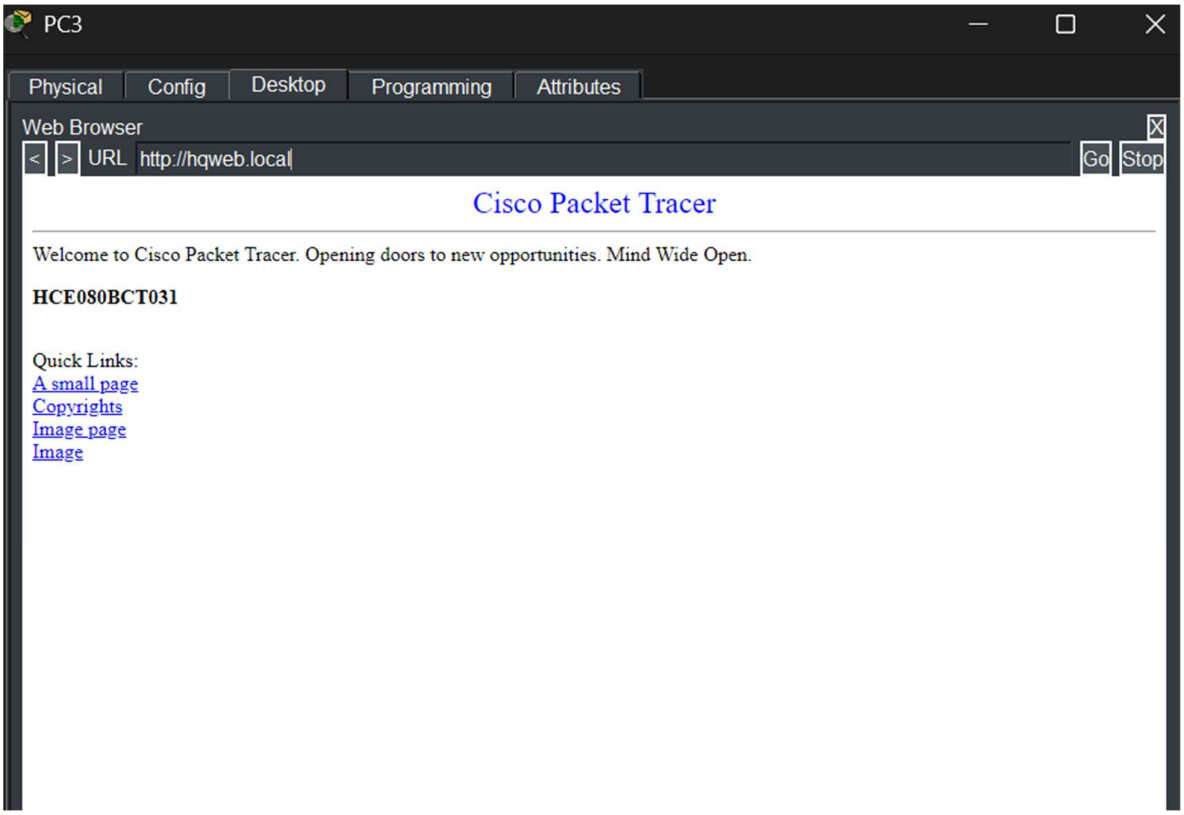
The Web server is located at Branch1 site with IP address 192.168.0.50. It hosts an HTTP web



10.3 DNS Resolution Test

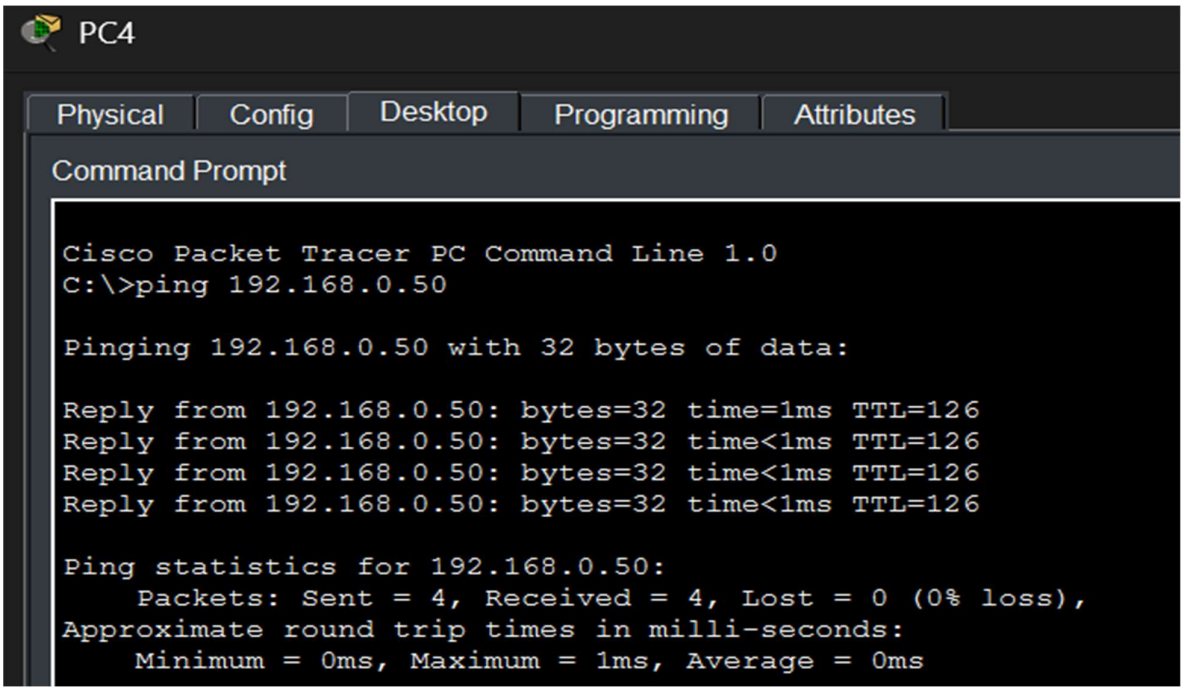
```
Cisco Packet Tracer PC Command Line 1.0
C:\>nslookup hqweb.local 192.168.0.60

Server: [192.168.0.60]
Address:  192.168.0.60

Non-authoritative answer:
Name:   hqweb.local
Address:   192.168.0.50
```

10.4 Web Browser Test

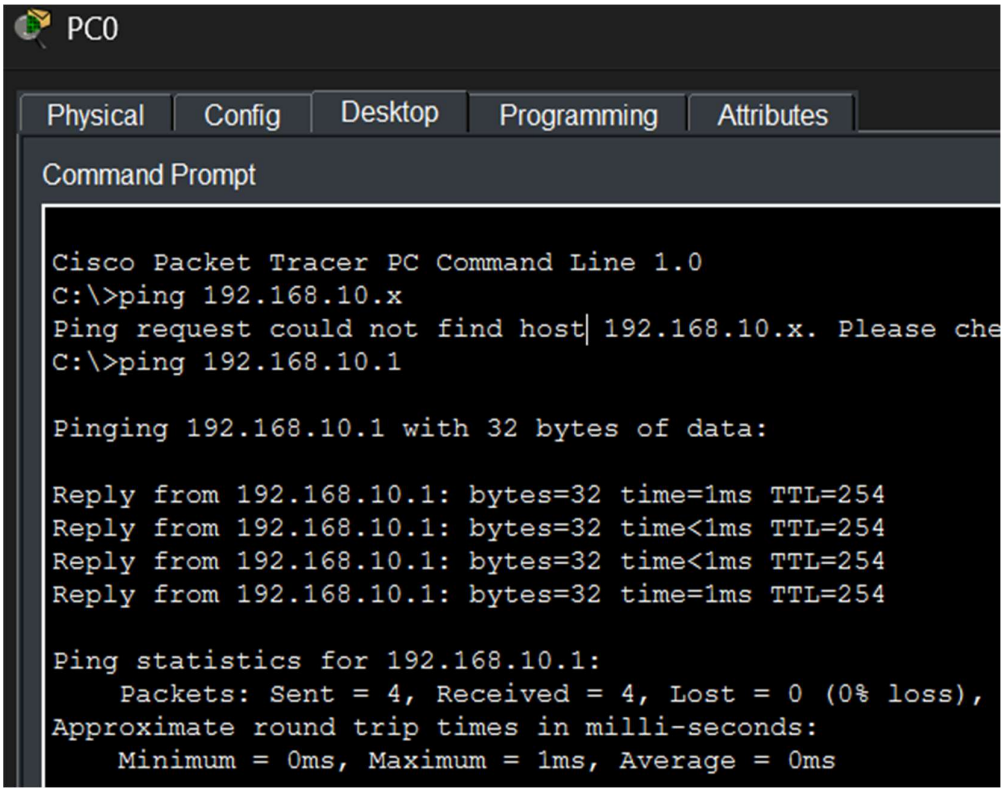


11. Verification and Testing

11.1 Ping Test - Branch to HQ



11.2 Ping Test - HQ to Branch



11.3 OSPF Neighbor Verification

```
Router#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           1   FULL/BDR        00:00:31    10.0.0.2        GigabitEthernet0/1
3.3.3.3           1   FULL/BDR        00:00:31    10.0.0.6        GigabitEthernet0/2
Router#
```

Test	Source	Destination	Expected Result
Ping HQ to Branch1	PC0 (192.168.0.x)	192.168.10.x	4/4 replies
Ping HQ to Branch2	PC0 (192.168.0.x)	192.168.20.x	4/4 replies
Ping Branch1 to HQ	PC2 (192.168.10.x)	192.168.0.50	4/4 replies
Ping Branch2 to HQ	PC4 (192.168.20.x)	192.168.0.50	4/4 replies

Test	Source	Destination	Expected Result
DNS Resolution	Any PC	hqweb.local	Resolves to 192.168.0.50
Web Access	Any PC Browser	hqweb.local	HTTP page loads

12. Troubleshooting - Routing Loop

12.1 Problem Identified

During the project, redistribution was accidentally configured on all three routers instead of only on HQ-R1. This caused a routing loop where:

1. Branch1-R2 was running both OSPF and RIP with redistribution
2. Branch2-R3 was running both OSPF and RIP with redistribution
3. Routes were being redistributed back into the protocol they originated from
4. HQ-R1 showed O E2 routes instead of R routes for Branch1 networks

12.2 Problem Screenshot

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/2
L       10.0.0.5/32 is directly connected, GigabitEthernet0/2
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.20
R    192.168.10.0/24 [90/1] via 10.0.0.2, 00:00:14, GigabitEthernet0/1
R    192.168.11.0/24 [90/1] via 10.0.0.2, 00:00:14, GigabitEthernet0/1
R    192.168.20.0/24 [90/5] via 10.0.0.6, 00:14:50, GigabitEthernet0/2
R    192.168.21.0/24 [90/5] via 10.0.0.6, 00:14:50, GigabitEthernet0/2
```

12.3 Solution Applied

The following steps were taken to resolve the routing loop:

1. Removed OSPF completely from Branch1-R2 (no router ospf 1)
2. Removed RIP completely from Branch2-R3 (no router rip)
3. Changed RIP administrative distance on R1 to 90 (distance 90) to prefer RIP over OSPF for Branch1 routes
4. Added passive-interface GigabitEthernet0/2 on R1 to stop RIP updates going to R3 side
5. Cleared ip route cache (clear ip route *) to flush stale routes

12.4 Fix Verified

```
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/1
L       10.0.0.1/32 is directly connected, GigabitEthernet0/1
C       10.0.0.4/30 is directly connected, GigabitEthernet0/2
L       10.0.0.5/32 is directly connected, GigabitEthernet0/2
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, GigabitEthernet0/0.10
L       192.168.0.1/32 is directly connected, GigabitEthernet0/0.10
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0.20
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0.20
R    192.168.10.0/24 [90/1] via 10.0.0.2, 00:00:01, GigabitEthernet0/1
R    192.168.11.0/24 [90/1] via 10.0.0.2, 00:00:01, GigabitEthernet0/1
O    192.168.20.0/24 [110/2] via 10.0.0.6, 00:00:12, GigabitEthernet0/2
O    192.168.21.0/24 [110/2] via 10.0.0.6, 00:00:12, GigabitEthernet0/2
```

13. Conclusion

This project successfully demonstrated the design and implementation of a multi-site branch-to-headquarters enterprise network using Cisco Packet Tracer. The following objectives were achieved:

Objective	Status
3-router topology with HQ and 2 Branch sites	Completed
VLAN configuration (VLAN 10 SALES, VLAN 20 IT) at each site	Completed
OSPF configured between HQ-R1 and Branch2-R3	Completed
RIP v2 configured between HQ-R1 and Branch1-R2	Completed
Route redistribution on HQ-R1 (OSPF <-> RIP)	Completed
DHCP configured on routers for dynamic IP assignment	Completed
DNS server resolving hqweb.local to web server IP	Completed
Web server accessible from all sites	Completed
Routing loop identified and troubleshot successfully	Completed
End-to-end connectivity verified with ping tests	Completed

The key learning from this project was understanding how route redistribution works at the boundary between two different routing protocol domains, and how incorrect redistribution on multiple routers can cause routing loops. The issue was resolved using passive interfaces and administrative distance adjustment.